Diffusion-induced stress and mechanical degradation in alloy-type nanoparticle anodes for high-performance lithium-ion batteries

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고성능 리튬이온 전지를 위한 합금계 나노입자 음극에서의 확산 유도 응력 및 기계적 열화

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Abstract

Alloy-type materials have attracted considerable attention as potential anodes for high-capacity lithium-ion batteries. However, their practical implementation is severely hindered by substantial volumetric expansion during cycling, which often results in pronounced capacity fading and degraded cycling stability. Understanding diffusion-induced stress is therefore essential for designing mechanically robust electrode architectures capable of long-term structural integrity. In this study, a chemo-mechanical model is employed to systematically investigate the evolution of lithiation-induced stress and strain in three representative alloy-type nanoparticles (Si, Ge, and Sn). A constitutive stress–strain relationship that explicitly incorporates concentration gradients is developed to accurately capture the stress response of the anodes during lithiation. Based on this framework, an elastic operating window is defined to identify optimal particle geometries corresponding to appropriate C-rate ranges. These results provide quantitative guidelines for material selection and particle size optimization tailored to specific requirements, thereby offering a rational design strategy toward high-energy-density and sustainable battery technologies.

Keywords: ***Alloy-type Anode, Chemo-mechanical model, High-capacity, Lithium-ion battery.***

1. Introduction

Lithium-ion batteries are rapidly approaching the performance limits of conventional intercalation anodes, prompting a shift toward high-capacity alloy-based materials like silicon (Si), germanium (Ge), and tin (Sn). However, their practical implementation is severely hindered by substantial volumetric expansion during cycling, causing severe capacity decay and compromised cycling performance. These mechanical stresses can induce cracking of the solid electrolyte interphase (SEI), leading to overall battery failure, including electrical contact loss, reduced capacity, higher impedance, and thermal disorder [1].

To overcome these barriers, nanostructuring strategies, particularly the use of nanoparticles, have been widely adopted. The proven efficacy of these architectures in enhancing electrochemical performance has positioned them as strong candidates for next-generation energy storage. Although solid nanoparticles have shown promise, experimental evidence suggests that hollow morphologies may better maintain the material within a stable elastic regime. Consequently, this study systematically compares solid and hollow nanoparticles based on their elastic behavior to identify the optimal design. This optimization requires balancing complex trade-offs: (i) smaller particles reduce stress but hinder scalability; (ii) thinner shells mitigate stress but compromise volumetric energy density; and (iii) material selection must support high C-rates (Fig. 1) [2].

2. Theoretical modeling

2.1 Chemo-mechanical modeling

The constitutive stress–strain relation accounting for concentration gradients is expressed as follows:

$$\frac{d^2u}{dr^2}+\frac{2}{r}\frac{du}{dr}-\frac{2u}{r^2}=\frac{(1+\nu)}{(1-\nu)}\frac{\Omega}{3}\frac{d\tilde{c}}{dr} \tag{1}$$

where u denotes the radial displacement, c is the Li^+ concentration, and $\tilde{c}$ represents the concentration at the stress-free state, Ω is the partial molar volume of solute, and ν is Poisson's ratio. The displacement field $u(r, t)$, expressed as a function of the Li concentration $c(r, t)$, satisfies the mass-conservation equation:

$$\frac{\partial c}{\partial t}+\frac{1}{r^2}\frac{\partial}{\partial r}\left(cr^2\frac{\partial u}{\partial t}\right)=D\left[(1+\theta c)\left(\frac{\partial^2 c}{\partial r^2}+\frac{2}{r}\frac{\partial c}{\partial r}\right)+\theta\left(\frac{\partial c}{\partial r}\right)^2\right] \tag{2}$$

where D is the lithium diffusion coefficient, R is the gas constant, and T is the operating temperature. At the inner surface, where the particle is not in contact with the electrolyte, the boundary condition is given by:

$$\frac{\partial c}{\partial r}(r=R_{in}(t),t)=0 \tag{3}$$

At the particle's outer surface, the mass flux J within the particle is balanced with the flux entering or leaving the particle through the surrounding electrolyte:

$$(1+\theta c)\frac{\partial c}{\partial r}(r=R_{out}(t),t)=I \tag{4}$$

Under galvanostatic conditions, the current density is determined by the applied C-rate, as follows:

$$i(t) = C_{max}(0)\frac{R_{out}^3 - R_{in}^3}{3R_{out}^2(t)}\frac{FC}{3600} \quad (5)$$

where $C_{max}(0)$ denotes the maximum stoichiometric concentration in the undeformed state. Under potentiostatic conditions, the current density follows the Butler–Volmer equation:

$$i(t) = -2i_0 \sinh\left(\frac{F\eta}{2RT}\right) \quad (6)$$

where i_0 is the exchange current density at the lithiation interface, driven by the concentration difference of lithium atoms at the particle surface, and η denotes the overpotential [2].

2.2 Diffusion-induced stress for solid and hollow particles

The temporal evolution of maximum von Mises stress within spherical solid particles exhibits a distinct stress peak at the onset of lithiation that increases monotonically with particle diameter (Fig. 2a). Comparative analysis indicates that Si exhibits significantly higher stress levels than Ge or Sn, a disparity driven by steep lithium concentration gradients and pronounced volumetric expansion intrinsic to the silicon lattice. The transition to thin-walled structures substantially reduces the stress disparity between large and small particles, suggesting that the dependence of mechanical stability on particle size is effectively suppressed in hollow configurations (Fig. 2b). A comparative analysis of geometric tuning reveals that the hollow architecture proves most advantageous for Si, which exhibits a marked expansion of the elastic safe-operating window, whereas Sn shows only marginal stress mitigation with decreasing shell thickness (Fig. 3). The dependency of volumetric capacity on particle dimensions highlights an inherent trade-off, where reducing shell thickness to alleviate stress inevitably degrades volumetric energy density. The necessity for a multivariate optimization strategy is underscored by the distinct performance regimes of each material, where silicon proves optimal for weight-sensitive nanoscale applications, while germanium offers the mechanical robustness required to sustain high C-rate operation in microscale systems.

3. Conclusion

This study elucidates the chemo-mechanical behavior of Si, Ge, and Sn anodes, demonstrating that diffusion-induced stress reaches its maximum at the onset of lithiation and is most severe in silicon due to its substantial volumetric expansion. Geometric optimization reveals that implementing hollow architectures substantially broadens the elastic operating window, particularly for larger particles, although this stress mitigation inevitably involves a trade-off with reduced volumetric capacity. Consequently, achieving durable high-capacity anodes requires a multivariate design strategy that carefully balances particle dimensions against competing demands between mechanical integrity and energy storage capacity.

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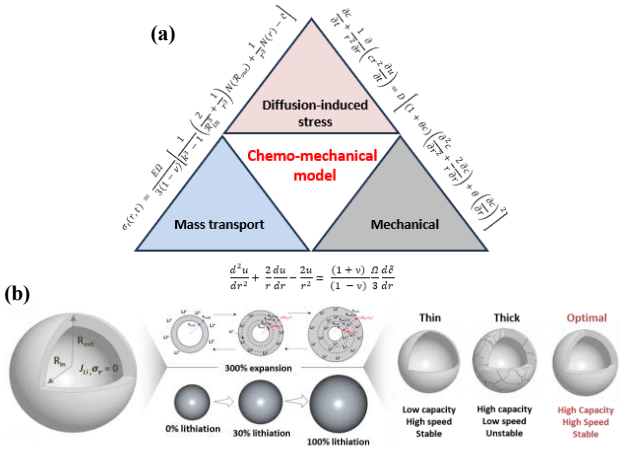

Fig. 1. (a) Chemo-mechanical formulation for analyzing the evolution of lithiation-induced stress in anode particles. (b) Geometric optimization for strain accommodation in high-capacity hollow anodes.

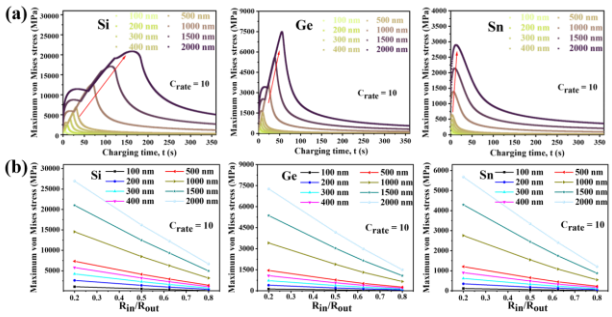

Fig. 2. (a) Temporal evolution of the maximum von Mises stress in a solid particle. (b) Maximum von Mises stress as a function of R_{in}/R_{out} for a hollow particle during the lithiation under galvanostatic conditions at 10C.

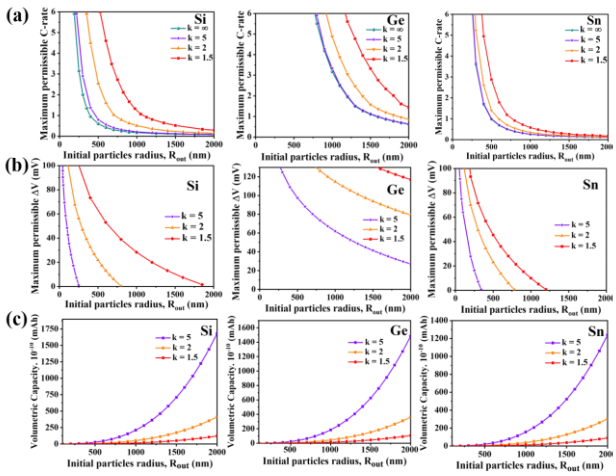

Fig. 3. Elastic-operation window for solid and hollow particles ($k = R_{out}/R_{in}$) of Si, Ge, and Sn under (a) galvanostatic and (b) potentiostatic conditions. (c) Influence of geometric parameters (R_{out} and k) on the volumetric capacity of hollow particles.